

RAFAEL ROHR

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EDUCATION

Duke University

Aug 2024 – May 2028 | Durham, NC (USA)

BS: Electrical & Computer Engineering | **Minors:** Finance, Math | **Relevant Coursework:** Probability, Financial Markets, Digital Systems, Computer Architecture
GPA: 3.9 | Dean's List with Distinction (Spring 2025), Dean's List (Fall 2025) | **Enrolled Classes:** Mathematical Finance (G), Adv. Computer Architecture I (G)

Honors and Awards: Biogenesis International Biomedicine Olympiad: Gold (2022, Molecular and Cellular Biology category); Gold (2022, Anatomy and Physiology category); Bronze (2022, Pharmacology and Toxicology category), Brazilian Chemistry Olympiad (**OBQ**): Bronze (2023); Bronze (2022), Brazilian Physics Olympiad (**OBP**): Silver (2023), Brazilian National Sciences Olympiad (**ONC**): Gold (2023); Gold (2022), Brazilian Astronomy and Astronautics Olympiad (**OBA**): Gold (2022), Brazilian Robotics Olympiad (**OBR**): Silver (2022); School's Outstanding Athlete 2020, 2021, and 2023; São Paulo's sports school's league's best goalkeeper

SKILLS & INTERESTS

Skills: Java, TypeScript, JavaScript, SQL, C, Python, Verilog, HTML, CSS, Onshape, Linux, API Development, Assembly

Technologies: Git, Vivado, FPGA, React, GTKWave, Postman, Docker Compose, Logisim, Arduino, Jira, Figma, Salesforce CRM, Tesseract Shell, Datadog

Languages: Portuguese (Native), English (C2), Spanish (C1), French (B1)

Interests: Soccer (Chelsea, Madrid, and Brazil), Chess, Philosophy (Epistemology and Metaphysics), Board and Card Games (7 Wonders, War, and Poker), Gym

INTERNSHIPS

StoneCo | *Software Engineer and Product Manager Intern*

May 2025 – Aug 2025 | Sao Paulo, SP (Brazil)

- Acted as PM and full-stack SWE in Brazil's largest fintech, building and deploying a feature that saved chat attendants time in each call by automating their workflow
- Saved 49 seconds per session (savings of ~R\$1M (US\$200K) annually) by implementing AI-driven prompt engineering with secure handling of sensitive user data
- Partnered with product, design, AI, and R&D teams to integrate Salesforce CRM, AWS S3, and Twilio to deliver a secure, scalable, and compliant fintech solution
- Created a fine-tuned ChatGPT model, conducted internal research to achieve accurate humanistic output, and used the custom model in a Typescript API endpoint

CAMPUS EXPERIENCE

Duke Annual Fund (Phonathon) | *Alumni Relations and Outreach Representative*

Jan 2026 - Present | Durham, NC (USA)

- Secured ~\$2,500 in contributions across 15+ fundraising sessions by building relationships with the Duke community through persuasive, high-impact communication
- Managed sensitive donor information, including payment and credit card data, with strict adherence to confidentiality, accuracy, and university data security protocols
- Developed stakeholder management, sales, and data-handling skills, engaging diverse alumni profiles in a fast-paced, metrics-driven outreach environment

Duke Department of Computer Science | *Undergraduate Teaching Assistant for Data Structure and Algorithms*

Aug 2025 – Present | Durham, NC (USA)

- Lead weekly class discussions for ~30 students, fostering critical thinking, reinforcing concepts in data structures, algorithms, recursion, and asymptotic analysis
- Hold office hours to individually support assignments, teaching debugging techniques, and guiding students to develop strong problem-solving and coding practices
- Assist with grading by providing thorough feedback on content, covering abstract data types, trees, graphs, and introductory AI/ML algorithms implemented in Java

PROJECTS & RESEARCH

Independent Study: Design for Testability (DFT) for pipelined RISC-V processor | *Researcher*

Jan 2026 – Present | Durham, NC (USA)

- Collaborated with Siemens EDA engineers to implement DFT methods using Tesseract Shell on a CPU, developing scan insertion, ATPG, and fault analysis scripts
- Achieved 95% stuck-at and 90.5% transition fault coverage, gaining expertise in digital design verification, semiconductor testing, and hardware reliability engineering

Spell-it-Out | *FPGA Engineer*

Apr 2026 – May 2026 | Durham, NC (USA)

- Built a whiteboard-writing robotic system integrating a self-designed Verilog 5-stage pipelined MIPS CPU, FPGA-controlled motors, servos, and PS2 keyboard input
- Developed assembly and memory-mapped FPGA interfaces for real-time text-to-motion execution, synchronized 3-axis control, and automated character formatting
- Debugged and optimized RTL, motor drivers, PWM servo control, and PS2 input processing to achieve reliable autonomous writing of all 26 alphabet characters

Christensen Family Center for Innovation | *Undergraduate Researcher*

Aug 2025 – May 2026 | Durham, NC (USA)

- Build an AI chatbot platform that streamlines founder ideation, opportunity evaluation, and venture planning through conversational AI and knowledge organization
- Research AI-driven methods to brainstorm alternative use cases for emerging engineering technologies, emphasizing innovation and commercialization potential
- Conduct market analysis and customer discovery to assess use-case feasibility, transforming qualitative feedback into product features and engineering requirements
- Explore MCAS (Multi-Camera Array Scanner) for Rapid On-Site Evaluation, enabling faster 3D imaging and AI-assisted diagnosis, transforming cytopathology

4Arms | *Chief Engineer*

Aug 2024 – Dec 2024 | Durham, NC (USA)

- Designed, constructed, and assembled with a spring mechanism, a 3D-printed fractured forearm prosthesis, so that residents learned to relocate a fractured forearm
- Tested the prototype with 8 physicians at the Duke Hospital, achieving a motion and force accuracy of 83%, establishing it as the hospital's official training equipment

LEADERSHIP & EXTRACURRICULAR

Brazilian Student Association (BRASA) | *Analyst, Vice-President, President*

Sep 2024 – Present | Durham, NC (USA)

- Plan and execute biweekly events, promoting a tight-knit Brazilian community and promoting our culture to foreign students, engaging 100+ students
- Secure over \$17k in funding, schedule cultural event logistics, and lead a team of 15 to execute programs that propelled student engagement successfully
- Developed professional partnerships with companies, such as EMS, Itaú, and StoneCo, and brought Alumni from Google, Microsoft, and Bain to info sessions
- Increased the number of members in 23%, and achieved 45,000 monthly accesses on our Instagram page, promoting BRASA beyond Duke's boundaries

Duke University Motorsports | *Powertrain Engineer*

Sep 2024 – May 2025 | Durham, NC (USA)

- Deep studied the powertrain system components of cars, better understanding its elements, such as the intake, the exhaust, the engine, the shifts, and the driveshaft
- Collaborated with team members to troubleshoot and resolve powertrain issues during testing and race events, install engines, and perform dyno tests